CSE4203: Computer Graphics

Lecture - 4 (Part - A)

Geometric Transformation

Outline

- Transformation
- Linear Transformation
  - Scaling
  - Shearing
  - Rotation
- Composite Transformation

2D Linear Transformations (1/1)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

Linear Transformation: Operation of taking a **vector** and produces **another vector** by a **simple matrix multiplication**.

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (1/6)

- The most basic transform is a scale along the coordinate axes.

$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

- S_x and S_y is called the scaling factors which determines how much scaling is applied along the x and y axis

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (2/6)

- The most basic transform is a scale along the coordinate axes.

$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

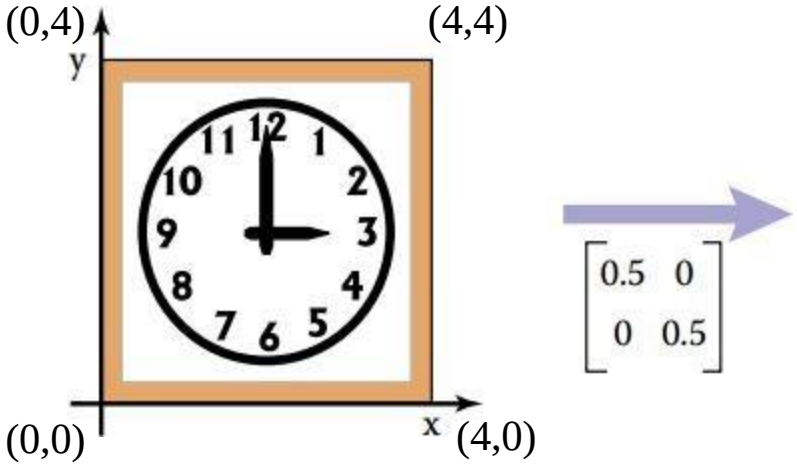
The matrix does to a vector with Cartesian components (x, y):

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

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Scaling (3/6)

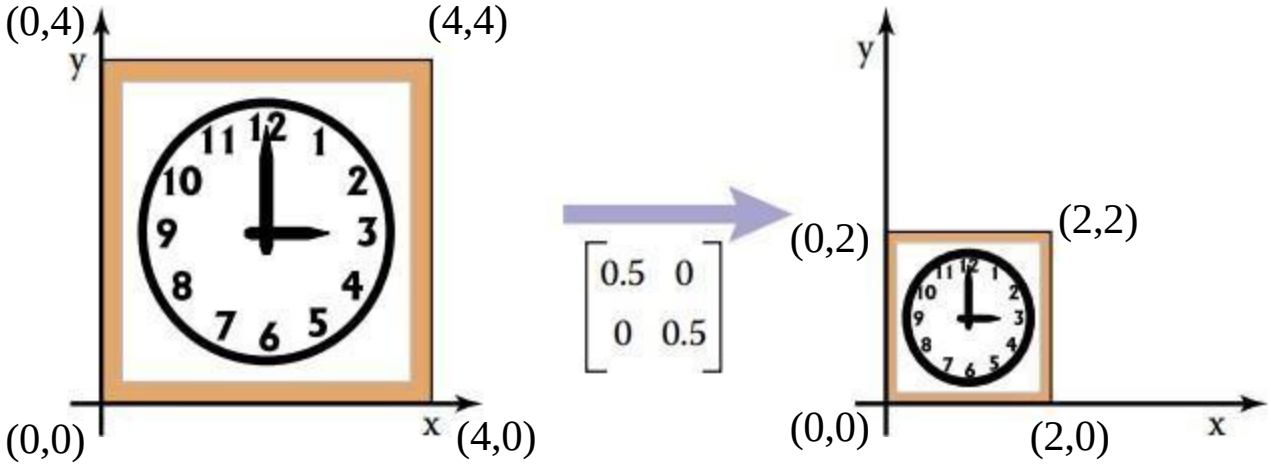
$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (4/6)

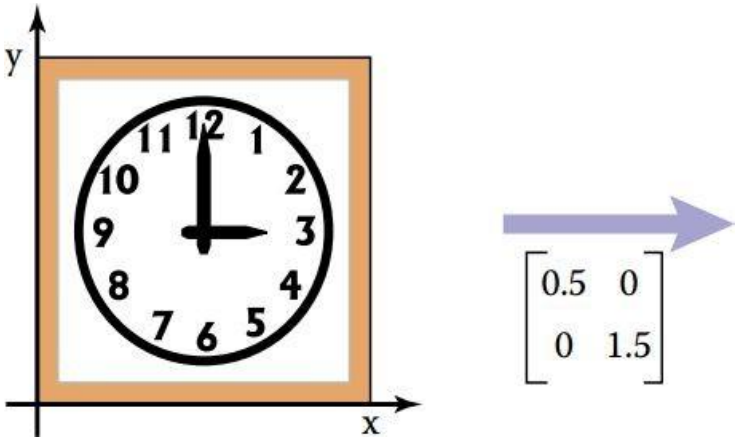
$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Scaling (5/6)

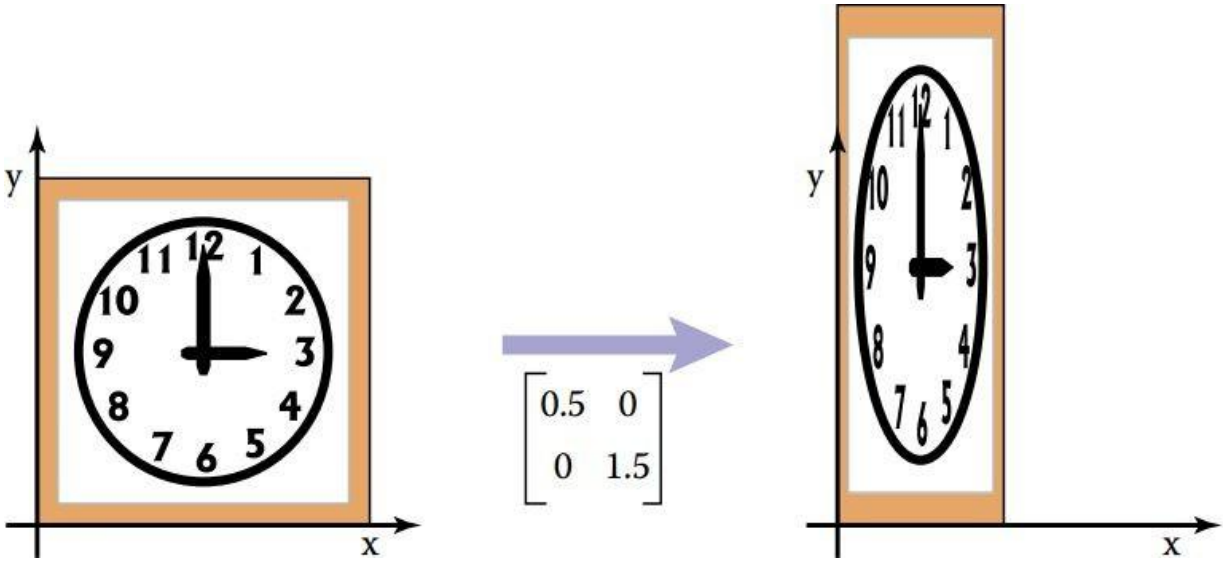
$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



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Scaling (6/6)

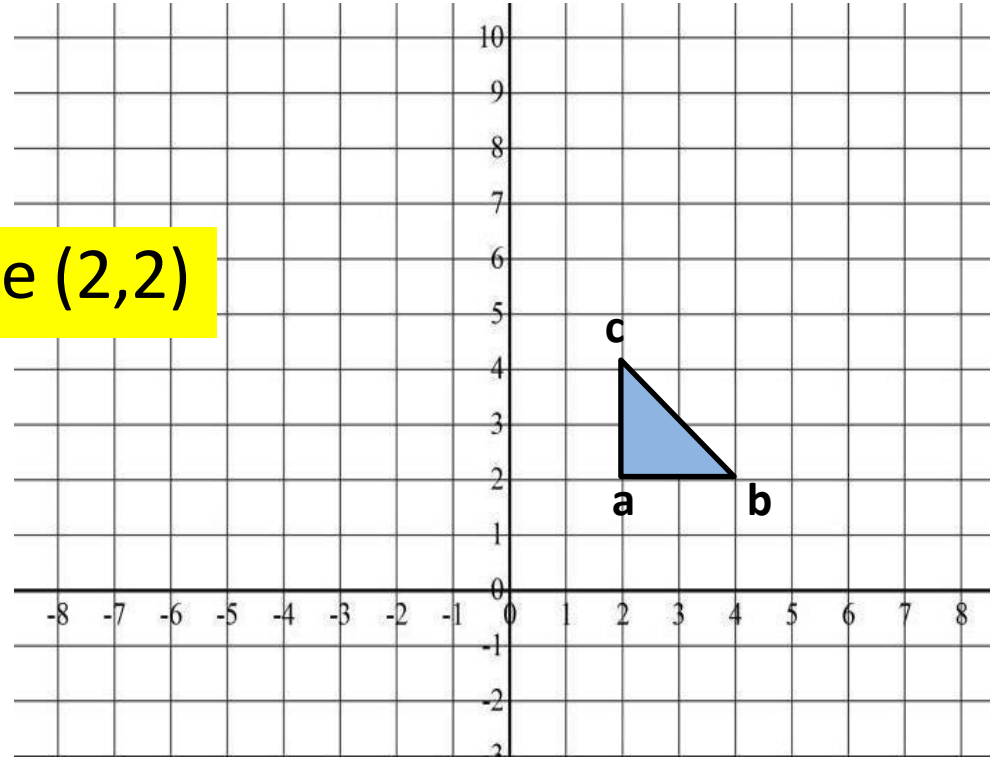
$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

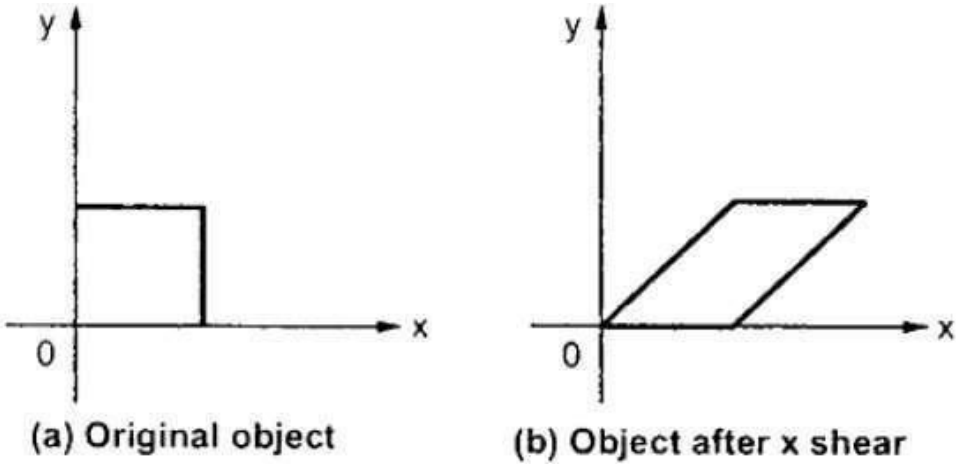
Scaling (6/6)

Scale (2,2)



Shearing (1/5)

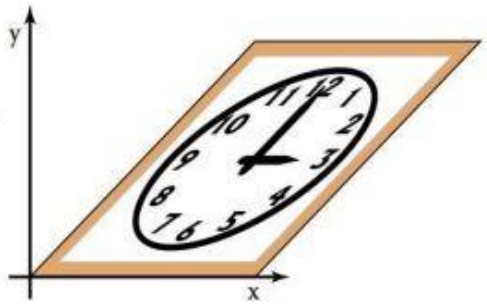
- A shear is something that pushes things sideways



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Shearing (2/5)

- A shear is something that pushes things sideways



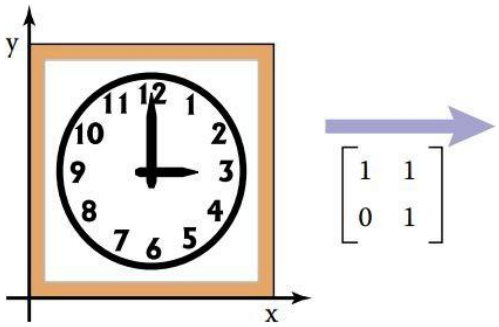


$$\text{shear-x}(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}, \quad \text{shear-y}(s) = \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix}$$

Source: https://www.tutorialspoint.com/computer_graphics/2d_transformation.htm

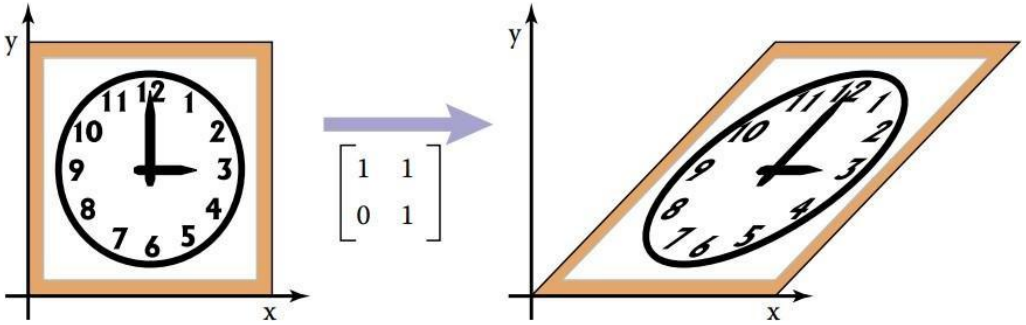
Shearing (3/5)

$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



Shearing (4/5)

$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

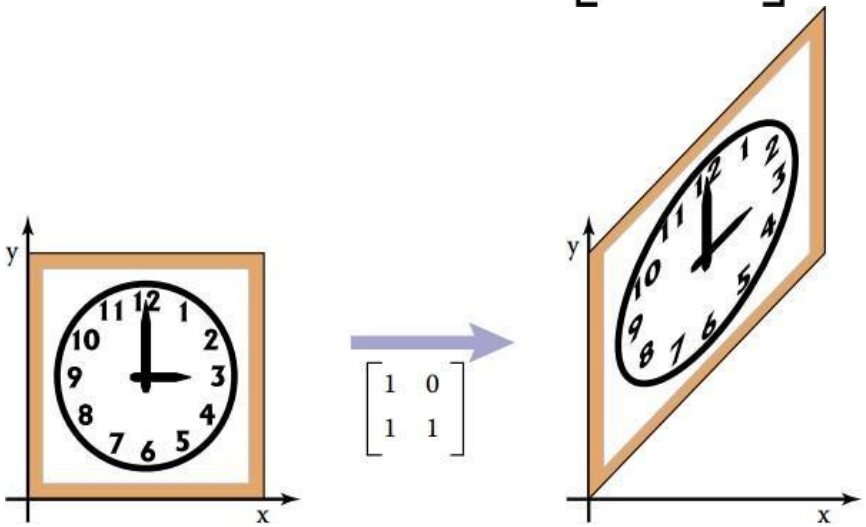


An x-shear matrix moves points to the right in proportion to their y-coordinate.

$$x - shear\ factor = \frac{displacement}{height}$$

Shearing (5/5)

$$\text{shear-y}(1) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$



A y-shear matrix moves points up in proportion to their x-coordinate

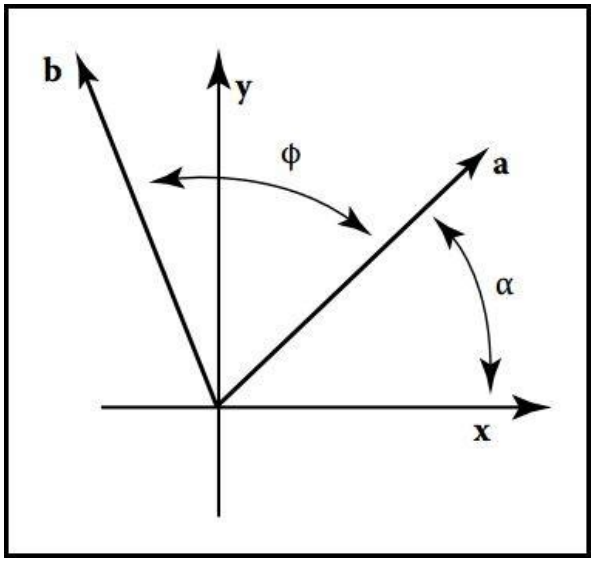
$$y - shear\ factor = \frac{displacement}{width}$$

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (1/11)

$$x_a = r\cos\alpha,$$

$$y_a =$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (2/11)

$$x_a = r\cos\alpha,$$

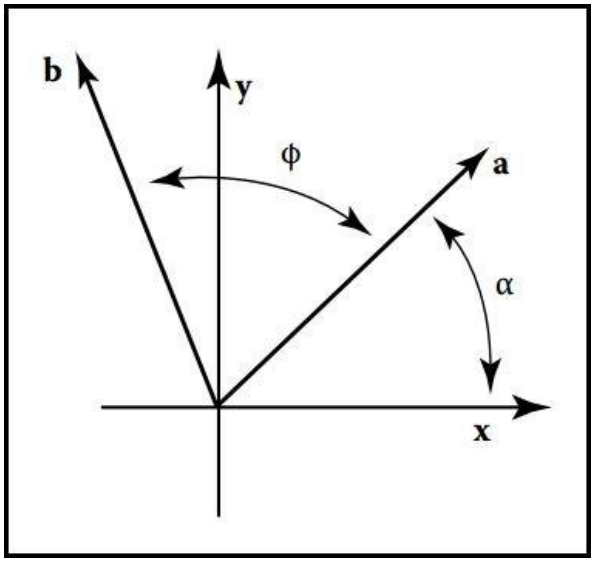
$$y_a = r\sin\alpha.$$

$$x_b = r\cos(\alpha+\phi) =$$

$$y_b =$$

Anti-clockwise rotation = (+) rotation

Clockwise rotation = (-) rotation



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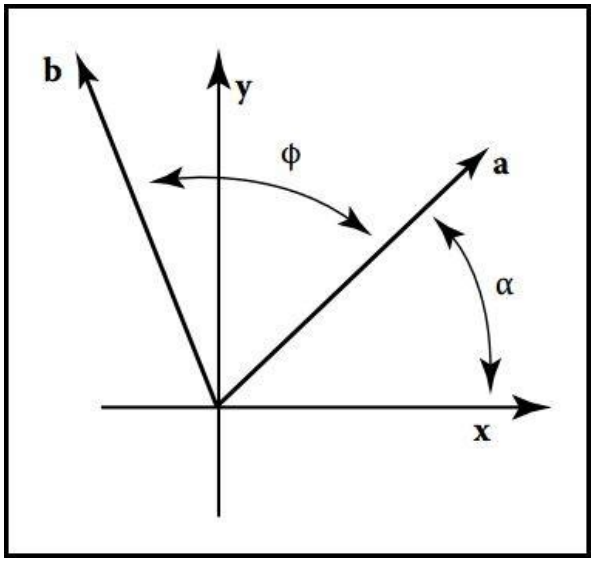
Rotation (3/11)

$$x_a = r\cos\alpha,$$

$$y_a = r\sin\alpha.$$

$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b =$$



Rotation (4/11)

$$x_a = r\cos\alpha,$$

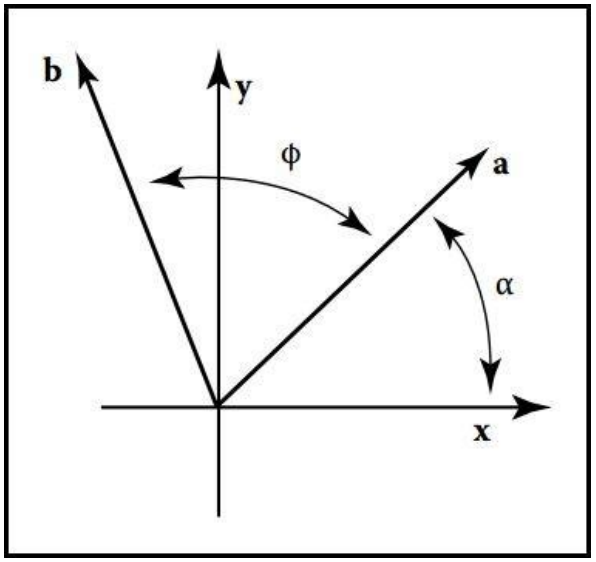
$$y_a = r\sin\alpha.$$

$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b = r\sin(\alpha+\phi) = r\sin\alpha\cos\phi + r\cos\alpha\sin\phi.$$

$$x_b =$$

$$y_b =$$



Rotation (5/11)

$$x_a = r\cos\alpha,$$

$$y_a = r\sin\alpha.$$

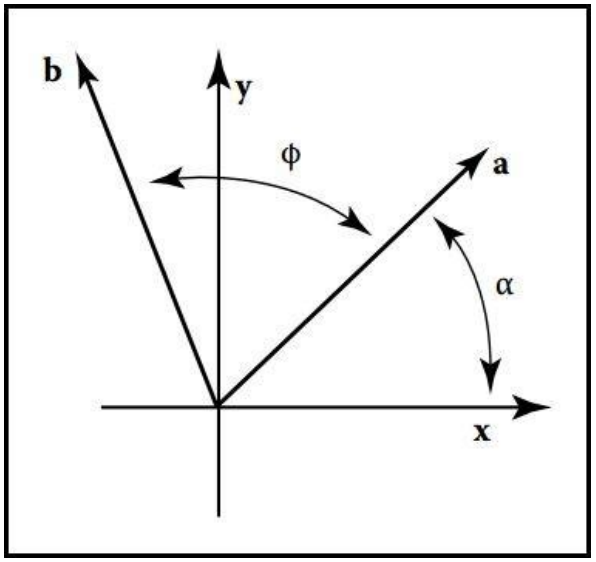
$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b = r\sin(\alpha+\phi) = r\sin\alpha\cos\phi + r\cos\alpha\sin\phi.$$

$$x_b = x_a\cos\phi - y_a\sin\phi,$$

$$y_b = y_a\cos\phi + x_a\sin\phi.$$

$$\text{rotate}(\phi) = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$



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Rotation (6/11)

$$x_a = r\cos\alpha,$$

$$y_a = r\sin\alpha.$$

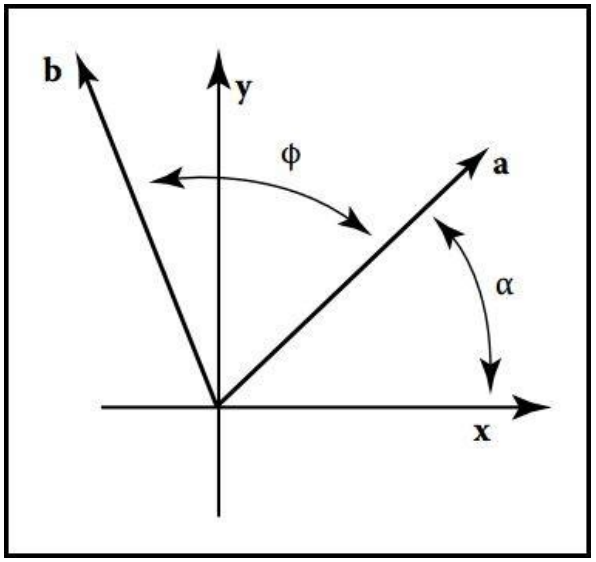
$$x_b = r\cos(\alpha+\phi) = r\cos\alpha\cos\phi - r\sin\alpha\sin\phi,$$

$$y_b = r\sin(\alpha+\phi) = r\sin\alpha\cos\phi + r\cos\alpha\sin\phi.$$

$$x_b = x_a\cos\phi - y_a\sin\phi,$$

$$y_b = y_a\cos\phi + x_a\sin\phi.$$

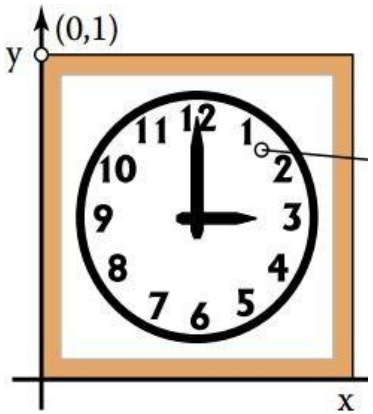
$$\text{rotate}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$



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Rotation (7/11)

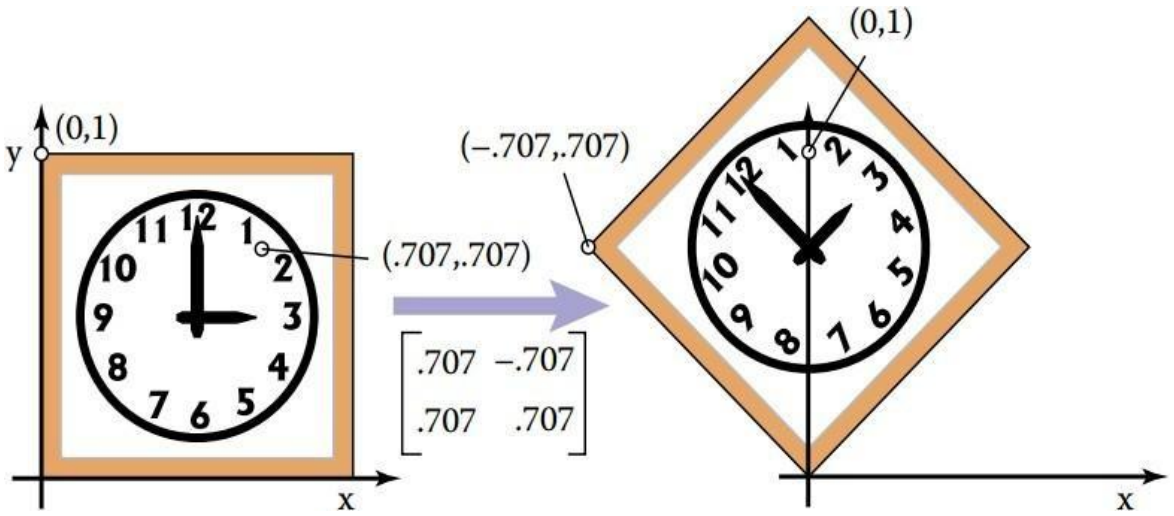
$$\begin{bmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



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Rotation (8/11)

$$\begin{bmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$

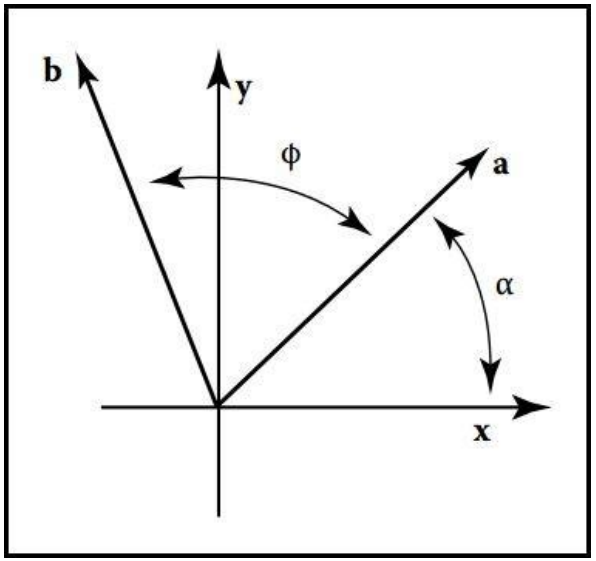


Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Rotation (9/11)

$$\text{rotate}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

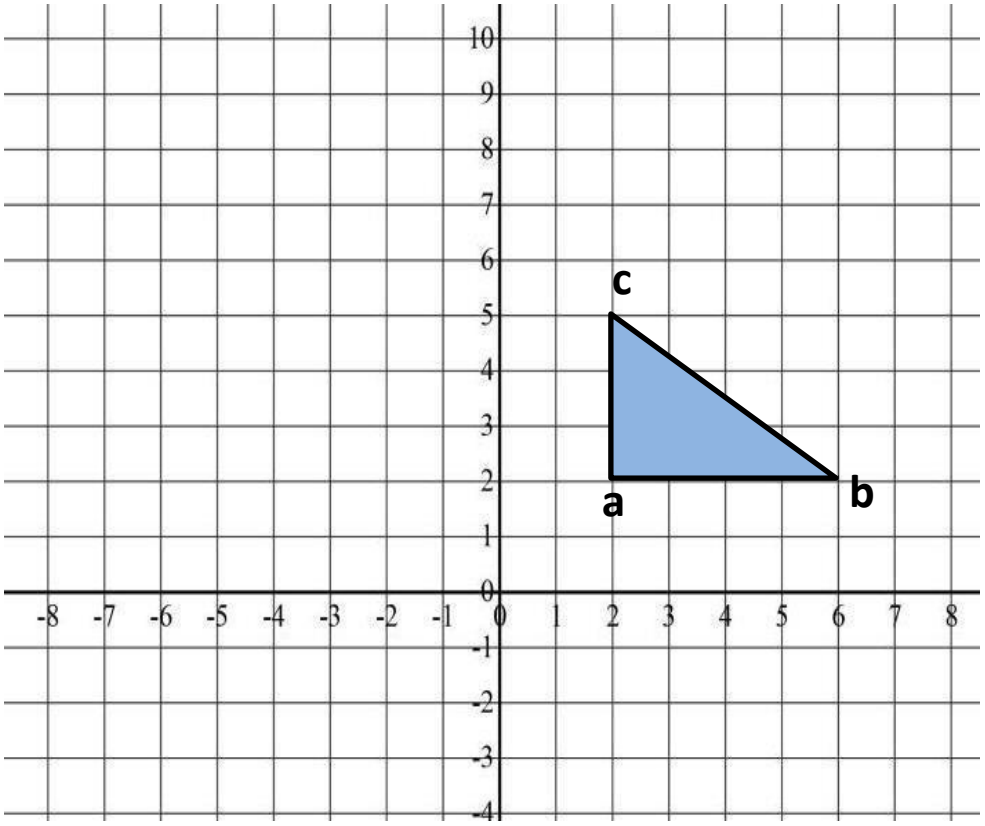
Q: What about – ve angle?



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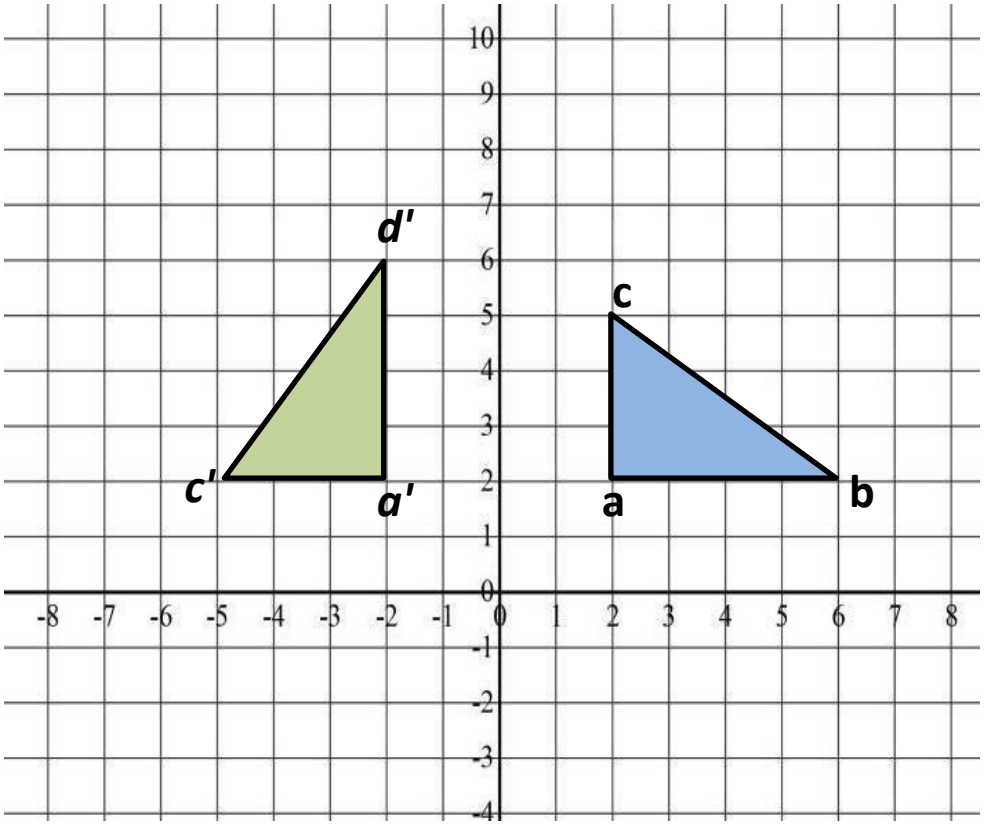
Rotation (10/11)

Rotate(90)



Rotation (11/11)

Rotate(90)



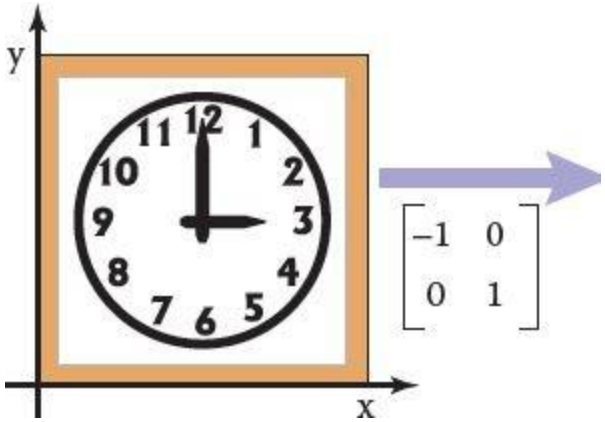
Reflection (1/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

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Reflection (2/5)

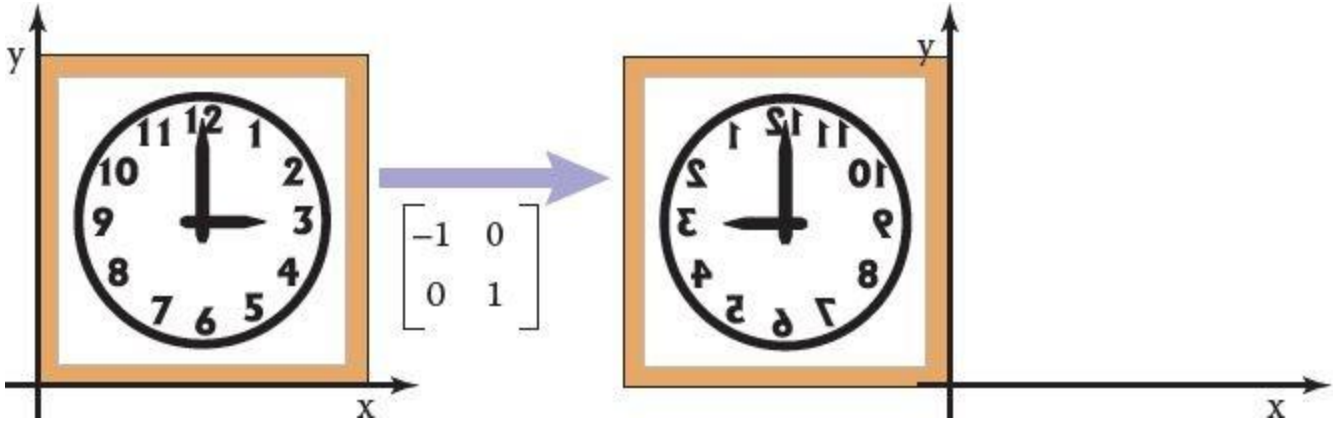
$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



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Reflection (3/5)

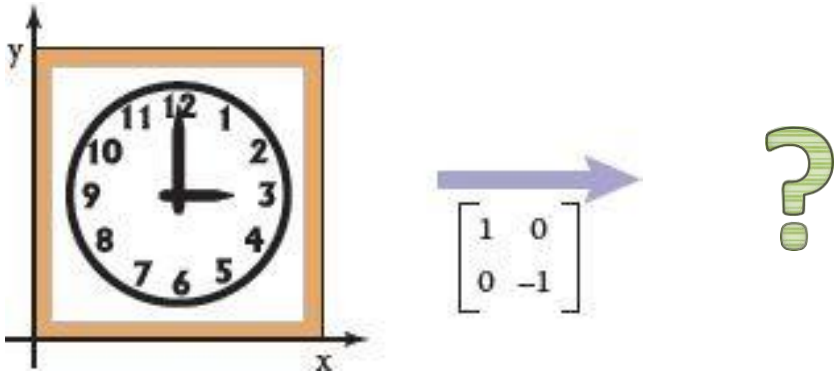
$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



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Reflection (5/5)

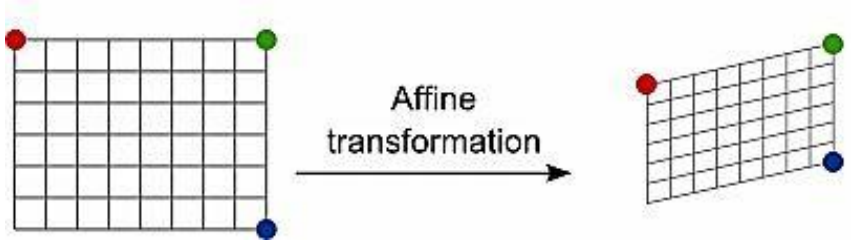
$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Affine transformation (1/2)

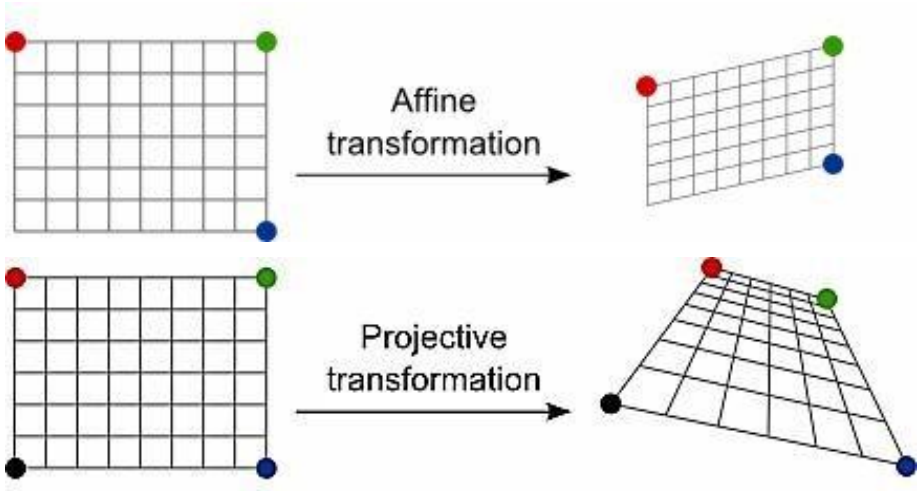
- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.



Source: https://www.graphicsmill.com/docs/gm/affine-and-projective-transformations.htm | https://en.wikipedia.org/wiki/Affine_transformation

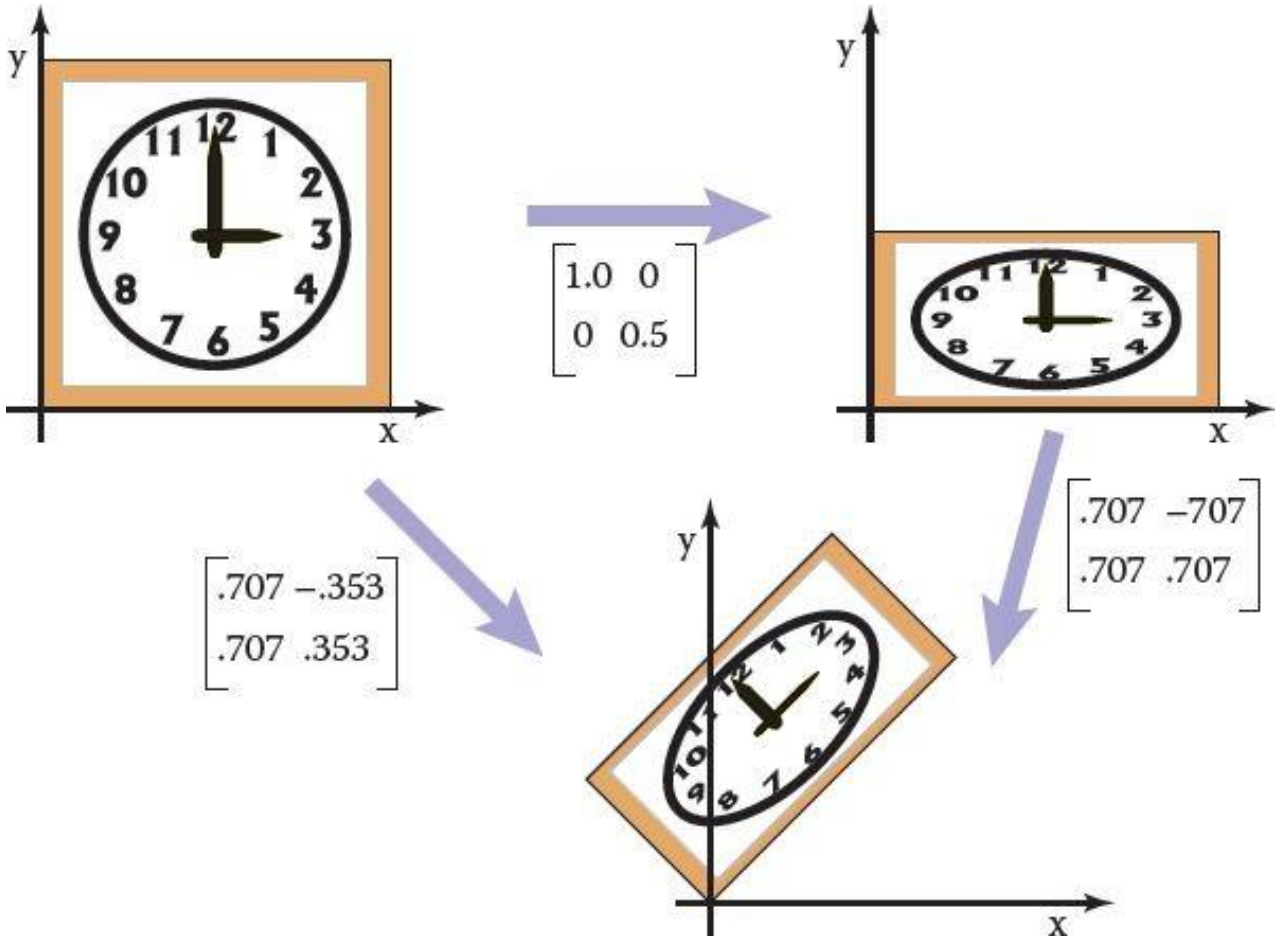
Affine transformation (2/2)

- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.



Source: https://www.graphicsmill.com/docs/gm/affine-and-projective-transformations.htm | https://en.wikipedia.org/wiki/Affine_transformation

Composition of Transformations (1/11)



Composition of Transformations (2/11)

- Composite transformation is the process of combining multiple transformations

  – i.e., for a 2D point v_1 we might want to –

  1. first apply a scale S
  2. then a rotation R.

- This would be done in two steps:

  1. **first, $v_2 = S\ v_1$**
  2. **then, $v_3 = R\ v_2$.**

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (3/11)

Therefore –

1. $v_2 = S\, v_1$
2. $v_3 = R\, v_2$
3. $v_3 = R\,(Sv_1)$
4. $v_3 = (RS)\, v_1$ *[matrix multiplication is associative]*
5. $v_3 = Mv_1$ *[Where M=RS]*

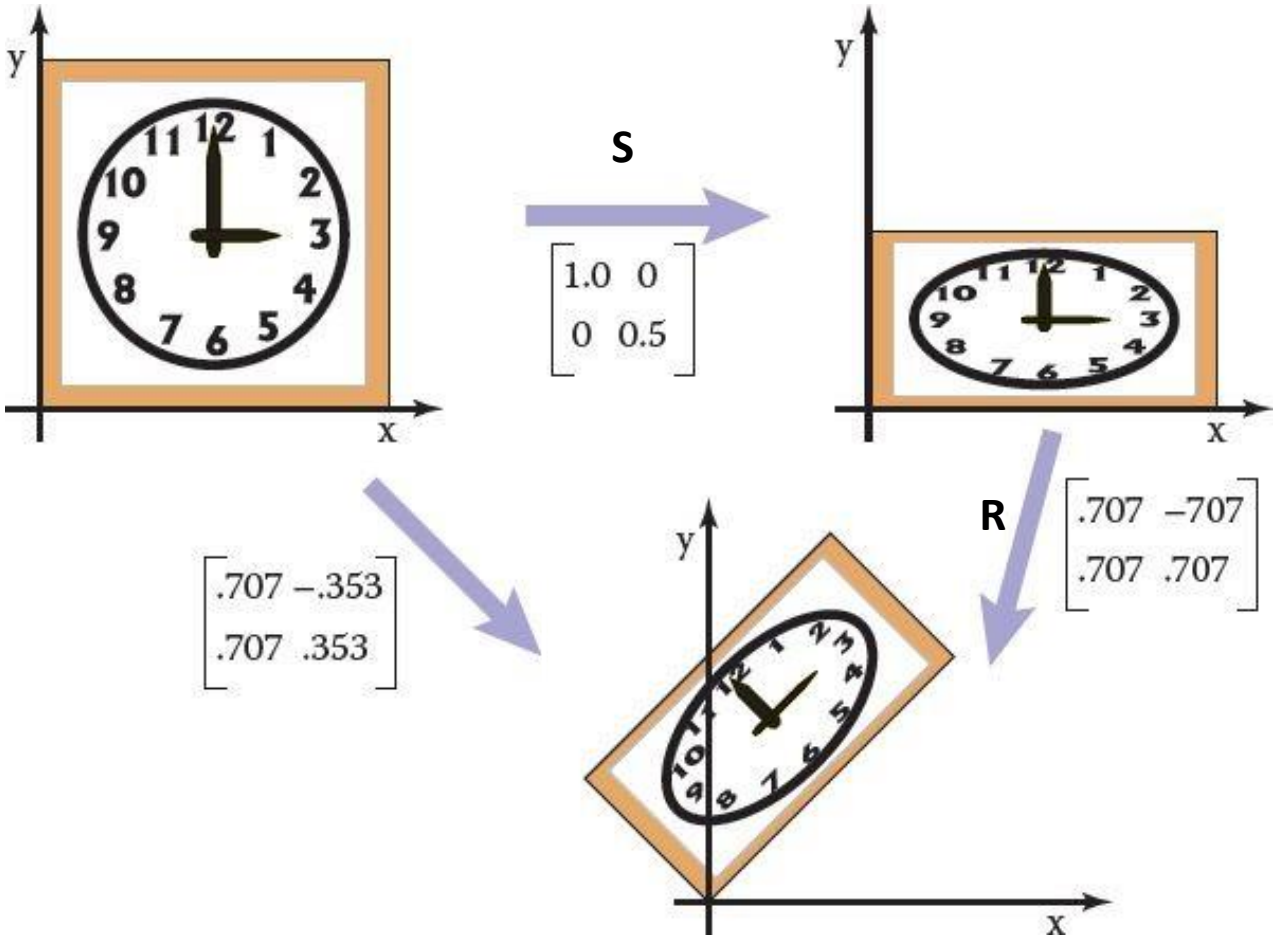
Composition of Transformations (4/11)

$$\mathbf{v}' = \mathbf{M}\,\mathbf{v}$$

[Where M = R S]

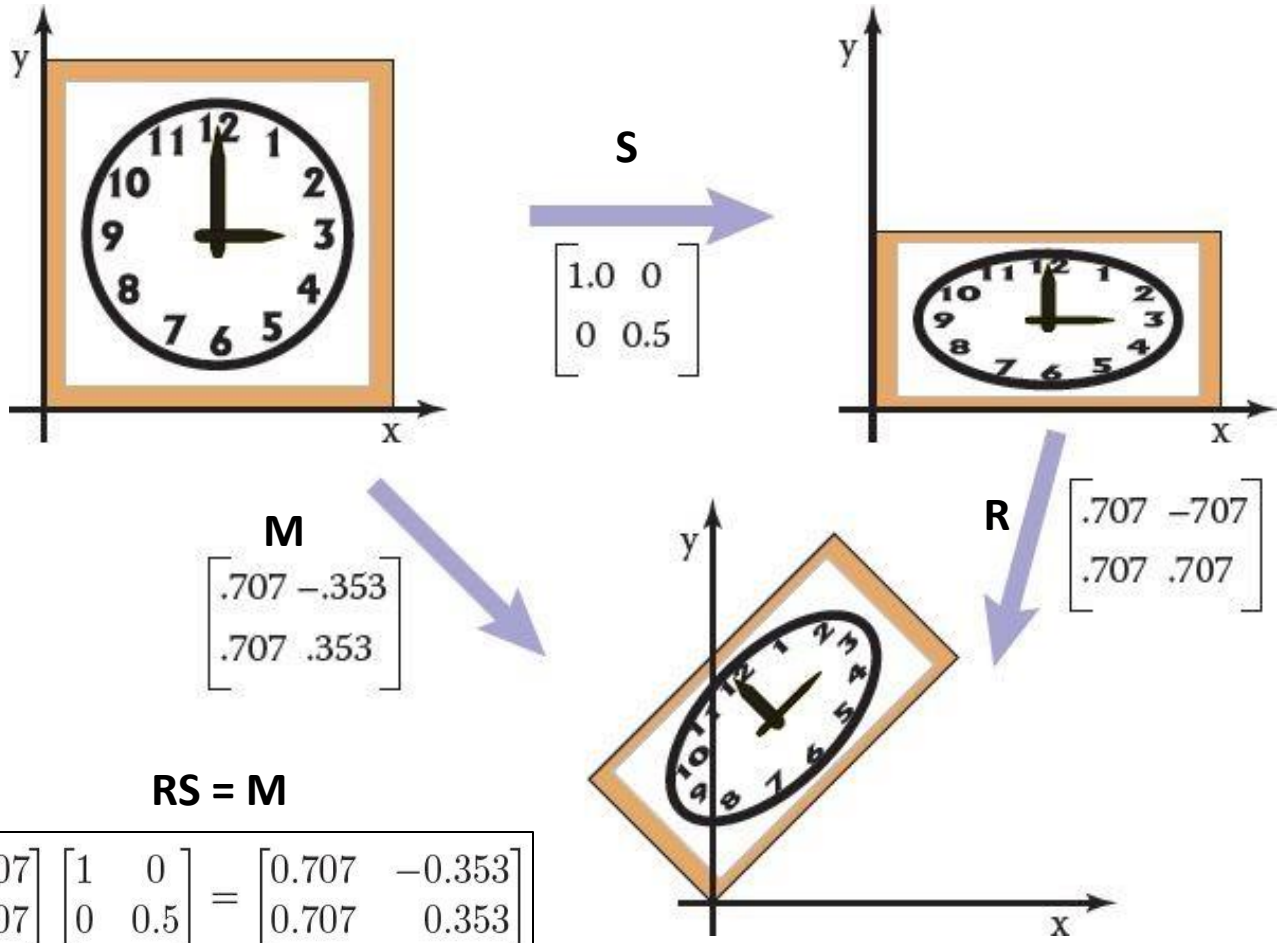
- We can represent the effects of transforming a vector by two matrices in sequence using a single matrix of the same size
  - which we can compute by multiplying the two matrices: **M = RS**

Composition of Transformations (6/11)



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (7/11)



RS = M

$$\begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.707 & -0.353 \\ 0.707 & 0.353 \end{bmatrix}$$

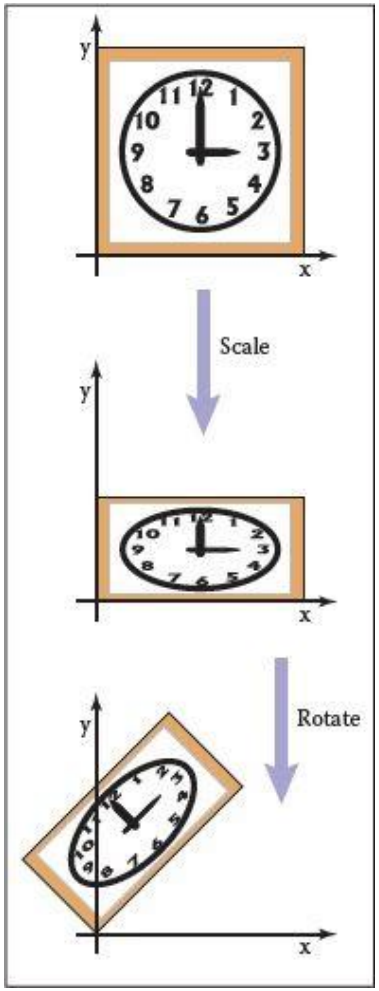
Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (8/11)

- It is very important to remember that these transforms are applied :
  - **from the right side first.**
  - So the matrix M= RS
    - first applies S and then R.

Composition of Transformations (9/11)

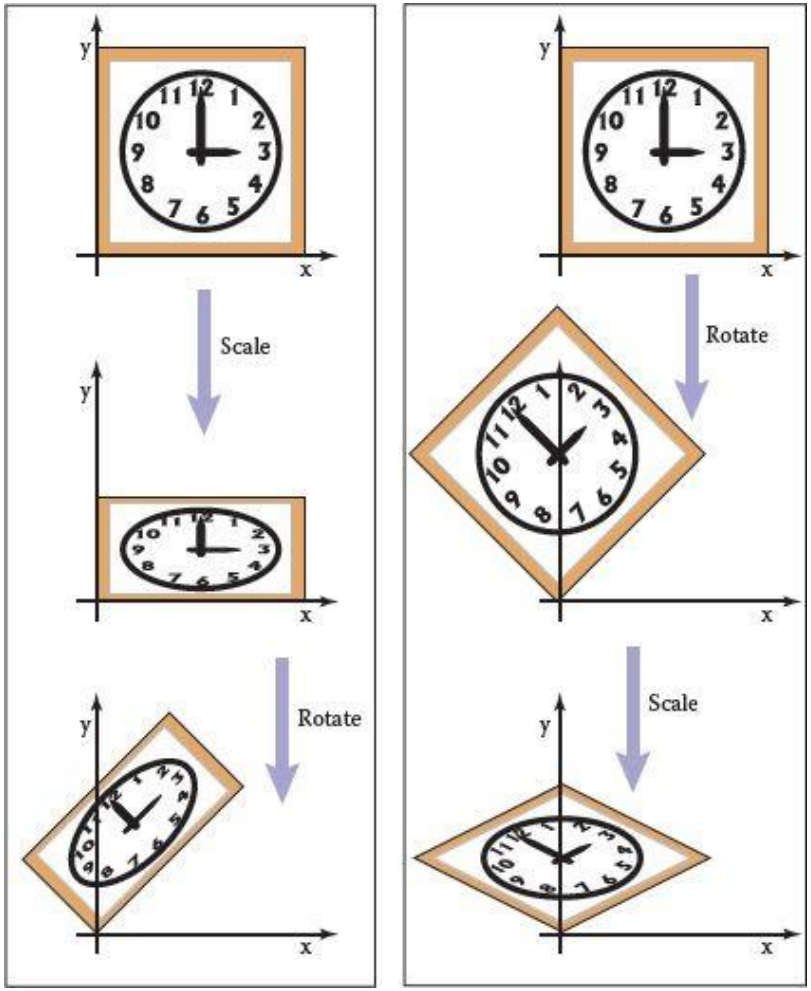
M= RS



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (10/11)

M= RS

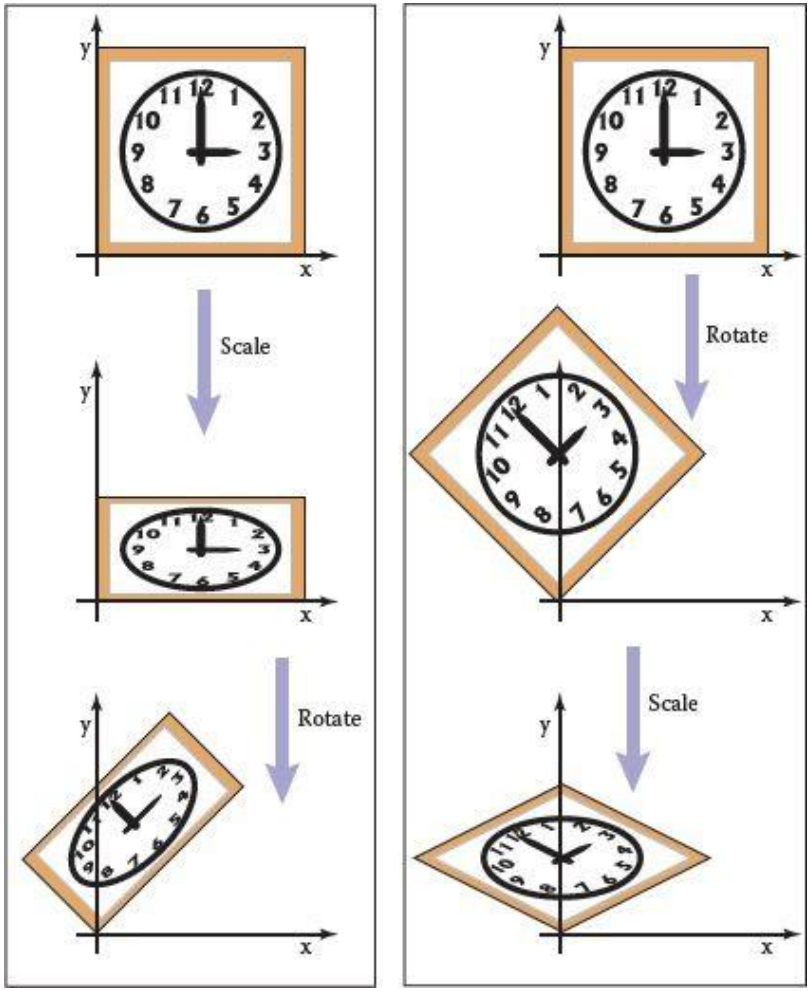


M = ?

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Composition of Transformations (11/11)

M= RS



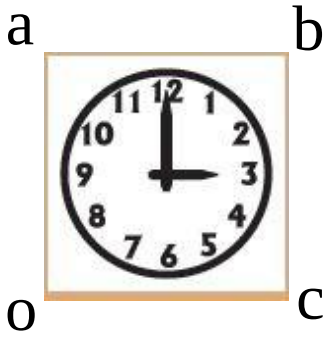
M=

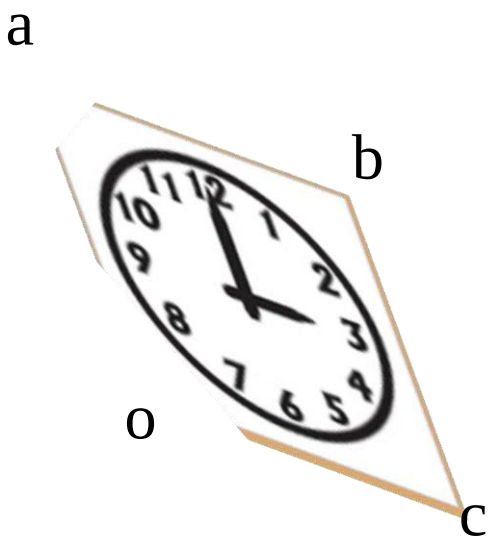
Q: What about more than two transformations: T1→T2→T3….→Tn

Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Practice Problem - 1

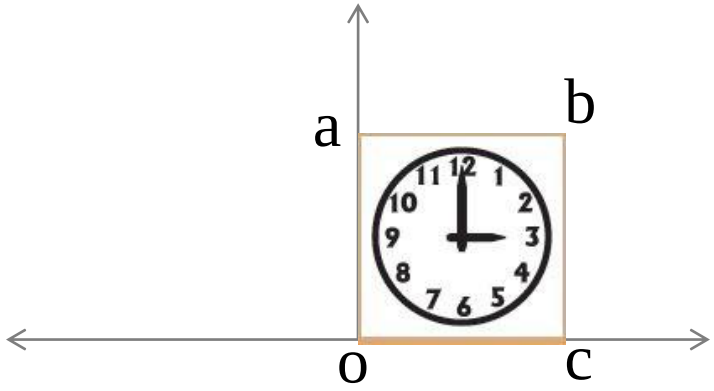
- Stretch the clock by 50% along one of its diagonals
  - so that 8:00 through 1:00 move to the northwest and 2:00 through 7:00 move to the southeast.



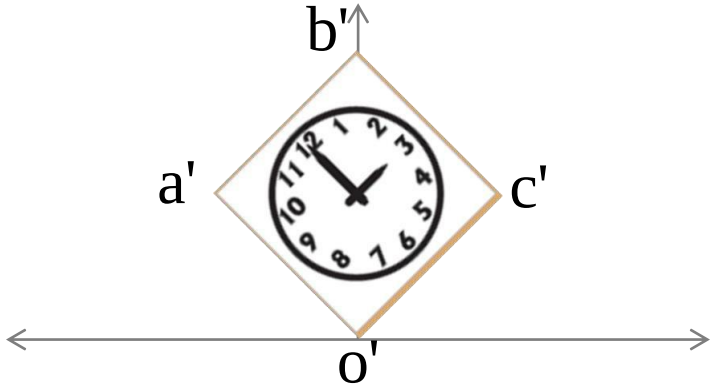


Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Practice Problem – 1 (Sol.)

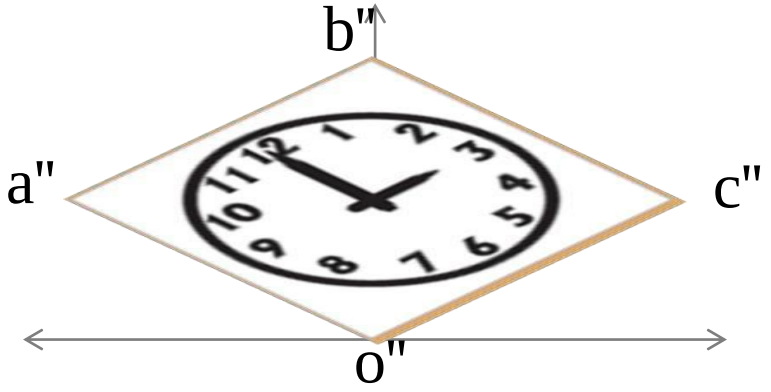


Practice Problem – 1 (Sol.)



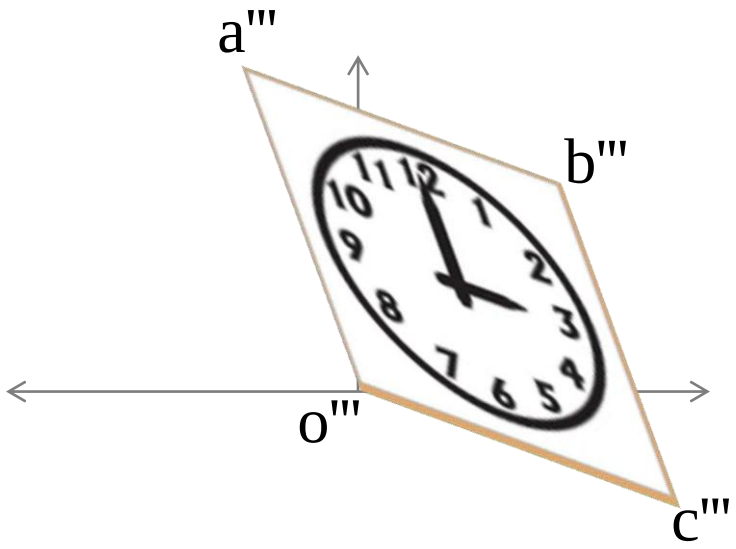
rotate(?)

Practice Problem – 1 (Sol.)



scale(?)

Practice Problem – 1 (Sol.)



rotate(?)

Practice Problem – 1 (Sol.)

- rotate(45°) → scale(1.5, 1) → rotate(− 45°).
  - *Q: Draw the steps*
- M = R(−45°) S(1.5, 1) R(45°)

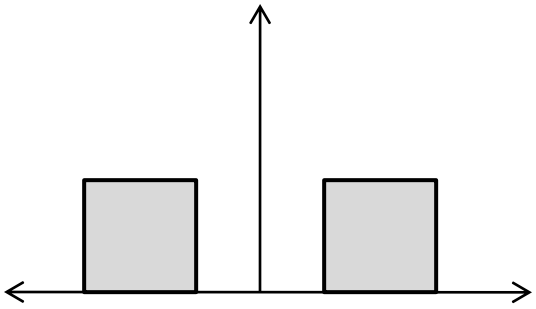
  $= R^T S R$
  - *Q: Calculate the matrix*

Practice Problem – 2

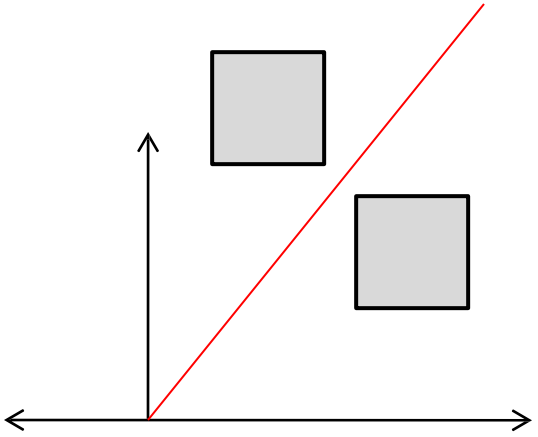
https://literary-peach-nwpto4rw.edgeone.dev/

- Reflect the clock along a line goes through origin:

$$y = mx + c$$

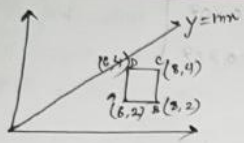


w.r.t y-axis



w.r.t arbitrary line

*



Reflect the clock along a line goes through the origin.

⇒ প্রথমে y=mx কে x অক্ষের y axis এর সাথে rotate করতে হবে।
তার জন্য ? ⇒ y = mx ⇒ y = x (m=1 হওয়ায়)

$\tan\theta = 1$
$\Rightarrow \theta = \tan^{-1}(1)$
$\Rightarrow \theta = 45^\circ$

⇒ R(45°)

then Reflect - y করতে হবে ⇒ Ref-y

then আবার R(-45) বা $R^T(45^\circ)$

⇒

$$V' = MV \quad \text{—①}$$

$$M = R(-45^\circ) \times \text{Ref-}y \times R(45^\circ) \quad \text{—②}$$

$$V = \begin{bmatrix} 6 & 8 & 8 & 6 \\ 2 & 2 & 4 & 4 \end{bmatrix}$$

$$R(-45^\circ) = \begin{bmatrix} \cos(45^\circ) & -\sin(45^\circ) \\ \sin(-45^\circ) & \cos(-45^\circ) \end{bmatrix} = \begin{bmatrix} 0.707 & 0.707 \\ -0.707 & 0.707 \end{bmatrix}$$

$$\text{Ref-}y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R(45^\circ) = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$

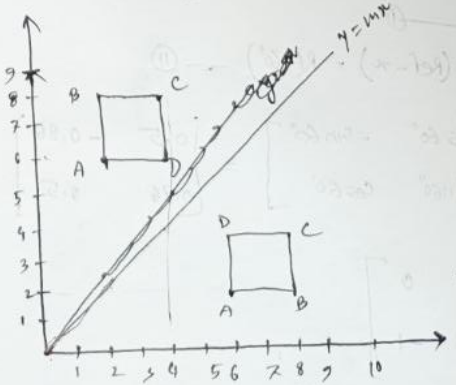
$$\therefore M = R(-45^\circ) \times \text{Ref-}y \times R(45^\circ)$$

$$= \begin{bmatrix} 0.707 & 0.707 \\ -0.707 & 0.707 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0.9996 \\ 0.9996 & 0 \end{bmatrix}$$

$$V' = MV = \begin{bmatrix} 0 & 0.9996 \\ 0.9996 & 0 \end{bmatrix} \begin{bmatrix} 6 & 8 & 8 & 6 \\ 2 & 2 & 4 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1.992 & 1.99 & 3.99 & 3.99 \\ 5.99 & 7.99 & 7.99 & 5.99 \end{bmatrix}$$



Q. Reflect the rectangle ABCD along the line of $y = \sqrt{3}x + 2$

$A(5,2),\ B(8,2),\ C(8,4),\ D(5,4)$

Line: slope $m = \sqrt{3} = \tan\theta \Rightarrow \theta = 60°,\ c = 2$

$c \neq 0$, so the line does not pass through origin → translate first.

Steps : $T(0,-2) \rightarrow R(-60°) \rightarrow \text{Ref-x} \rightarrow R(60°) \rightarrow T(0,2)$

$V' = M \cdot V$ ..... (I)

$M = T(0,2) \times R(60°) \times \text{Ref-x} \times R(-60°) \times T(0,-2)$ ..... (II)

$$T(0,-2) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \qquad T(0,2) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R(60°) = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad R(-60°) = \begin{bmatrix} 0.5 & 0.866 & 0 \\ -0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Ref-x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Finding M :

$P1 = R(-60°) \times T(0,-2)$

$$\begin{bmatrix} 0.5 & 0.866 & -1.732 \\ -0.866 & 0.5 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$P2 = \text{Ref-x} \times P1$

$$\begin{bmatrix} 0.5 & 0.866 & -1.732 \\ 0.866 & -0.5 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$P3 = R(60°) \times P2$

$$\begin{bmatrix} -0.5 & 0.866 & -1.732 \\ 0.866 & 0.5 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$M = T(0,2) \times P3$

$$\begin{bmatrix} -0.5 & 0.866 & -1.732 \\ 0.866 & 0.5 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

From (I): $V' = M \cdot V$

$$V' = \begin{bmatrix} -0.5 & 0.866 & -1.732 \\ 0.866 & 0.5 & 1 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & 8 & 8 & 5 \\ 2 & 2 & 4 & 4 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2.5 & -4 & -2.268 & -0.768 \\ 6.33 & 8.928 & 9.928 & 7.33 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

∴ Answer :

A'(−2.5, 6.33) B'(−4, 8.928)

C'(−2.268, 9.928) D'(−0.768, 7.33)

Further Reading

- Fundamentals of Computer Graphics, 4th Edition - Chapter 6

Thank you